

AAKASH REDDY NUTHALAPATI

Irving, TX | aakashreddy3702@gmail.com | +1-940-326-1786 | [linkedin.com/in/aakash-reddy-nuthalapati](https://www.linkedin.com/in/aakash-reddy-nuthalapati) | github.com/aakash-73

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2+ years of professional experience designing, building, testing, and deploying production-grade applications across the full stack. Proficient in Java, Python, JavaScript, TypeScript, React, React Native, Spring Boot, FastAPI, and AWS. Experienced building distributed systems, event-driven microservices (Apache Kafka), RESTful APIs, and AI-powered applications including RAG pipelines, LLM integrations, and vector search. Hands-on with Docker, Kubernetes, CI/CD pipelines, and infrastructure as code (AWS CDK, AWS CLI). M.S. Computer Science, University of Central Oklahoma. Currently working at Digitronix AI Inc.; open to full-time Software Engineer roles with visa sponsorship.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Kotlin, Golang (learning)

Backend: Java Spring Boot, Python FastAPI, Python Flask, REST APIs, Microservices Architecture, Event-Driven Architecture, Apache Kafka, Node.js

Frontend: React 18, React Native, React Flow, JavaScript (ES6+), TypeScript, HTML5, CSS3, Server-Sent Events (SSE), Responsive UI

AI and ML: RAG Pipelines, Vector Embeddings, HuggingFace Sentence Transformers, LLMs (Groq LLaMA3, Ollama, qwen2.5-coder), Prompt Engineering, AST Parsing, scikit-learn, Claude Code, GitHub Copilot

Cloud and DevOps: AWS (Lambda, API Gateway, DynamoDB, Cognito, Rekognition, Textract, CloudWatch, S3), AWS CDK, AWS CLI (IaC), Docker, Kubernetes, CI/CD (GitHub Actions)

Databases: MongoDB Atlas (NoSQL), MySQL (Relational/SQL), DynamoDB, Apache Ignite

Data Structures and Algorithms: Arrays, Trees, Graphs, Hash Maps, Dynamic Programming, BFS/DFS, Topological Sort (Kahn's Algorithm), Sorting and Searching, OOP Design Patterns

SDLC and Process: Agile/Scrum, TDD, Code Reviews, Technical Documentation, Version Control (Git/GitHub), Observability (CloudWatch)

Testing: JUnit 5, Mockito, Unit Testing, Integration Testing, ROUGE-L Evaluation

PROFESSIONAL EXPERIENCE

Full Stack Developer | Digitronix AI Inc. | May 2026 - Present

- Building AI-powered full stack applications at an AI-focused technology company using Python, React, and AWS; integrating large language models and cloud-based AI tools into production workflows to deliver intelligent automation capabilities for enterprise use cases.

Full Stack Developer | Saayam For All | Aug 2025 - Apr 2026

- Led a frontend engineering team of 5 developers in an Agile/Scrum environment across 3 production release cycles; conducted weekly code reviews enforcing SOLID principles and React Native best practices; mentored junior engineers through pair programming and structured technical walkthroughs.
- Architected and delivered a reusable React Native component library improving application performance by 25% and team development velocity by 30% across 4 core product modules.
- Integrated AWS Cognito for OAuth-based authentication and designed RESTful API contracts between frontend and backend microservices; resolved critical performance defects through log analysis and debugging, reducing page load time by 35%.
- Authored test plans and integration tests; used Claude Code and GitHub Copilot daily to accelerate development, documentation, and code quality across the codebase.

Machine Learning Engineer Intern | Techionary | May 2020 - Jul 2020

- Trained supervised and unsupervised ML models using Python (scikit-learn), improving model accuracy by 20% through feature engineering and hyperparameter tuning.
- Built end-to-end data preprocessing pipelines using Pandas and NumPy, cutting data preparation time by 35% and improving pipeline reliability for production ML workflows.

FEATURED PROJECTS

Smart Expense Tracker - Distributed Microservices System | 2024

Java, Spring Boot, MySQL, MongoDB, Apache Kafka, Docker, Kubernetes, Apache Ignite, React, GitHub Actions

- Architected a 6-service distributed system in Java using Spring Boot microservices with relational (MySQL) and NoSQL (MongoDB) databases; implemented event-driven inter-service communication via Apache Kafka, reducing inter-service latency by 40%.

- Containerized with Docker and orchestrated on Kubernetes with zero-downtime rolling updates; implemented distributed caching (Apache Ignite) reducing database query time by 50%; CI/CD pipeline (GitHub Actions) enforced 80%+ code coverage (JUnit 5, Mockito) before every production release.

RepoScope AI - Code Architecture Intelligence Platform | 2024 - 2025

Python, FastAPI, React 18, React Flow, MongoDB, Ollama, Claude Code, Server-Sent Events (SSE)

- Designed and shipped a full-stack AI-powered platform that ingests GitHub repositories and renders real-time dependency graphs; integrated a local LLM (Ollama + qwen2.5-coder) eliminating cloud API costs and data privacy risk.
- Implemented advanced algorithms (topological sort via Kahn's algorithm, DFS/BFS graph traversal) for dependency analysis; built an MD5-based incremental sync reducing re-analysis time by 80% and a real-time SSE streaming pipeline eliminating client polling overhead.

Syllabus Chatbot - RAG-Powered AI Application | 2024 - 2025

Python, Flask, React, MongoDB Atlas, HuggingFace Sentence Transformers, Groq LLaMA3, PyMuPDF

- Built a production RAG pipeline enabling natural language queries over PDF documents; achieved 100% retrieval hit rate and 62% ROUGE-L generation accuracy against ground-truth benchmarks using HuggingFace embeddings and MongoDB Atlas vector search.
- Designed a Python Flask REST API with primary/secondary Groq LLaMA3 failover ensuring zero-downtime AI inference; applied Chain-of-Thought and Few-Shot prompt engineering to maximize LLM response accuracy.

Serverless Participation Verification System | 2024

AWS Lambda, API Gateway, DynamoDB, Rekognition, Textract, CloudWatch, AWS CDK, AWS CLI

- Engineered a fully serverless event-driven system on AWS (Lambda, API Gateway, DynamoDB) provisioned with AWS CDK and AWS CLI (infrastructure as code); integrated Rekognition and Textract for AI-powered identity verification, reducing manual processing time by 70%.
- Implemented IAM role-based access control enforcing least-privilege security; CloudWatch monitoring provided real-time observability into production health.

Attendance Management System - Android | 2023

Kotlin, Jetpack Compose, MVVM, Hilt, Room

- Built a production Android application using Kotlin and Jetpack Compose with MVVM architecture; offline-first Room data sync ensured 100% data availability without network connectivity; Hilt dependency injection improved testability and reduced boilerplate by 40%.

EDUCATION

M.S. in Computer Science | University of Central Oklahoma | 2023 - 2025

Relevant Coursework: Data Structures and Algorithms, Distributed Systems, Cloud Computing, Machine Learning, Database Management, Software Engineering, Operating Systems

B.Tech in Electrical and Electronics Engineering | JNTU Hyderabad | 2019 - 2023